A transitive witness and its transported orbit

Human Compression project: L044 continuation

Let $G \leq \mathrm{Alt}(\Omega)$ be transitive with connected string C-marking $\rho_0, \ldots, \rho_{r-1}$, all maximal omissions intransitive, $r \geq 6$ and $n = 2r + 1$. The first three marks are supported on, and transitive on, the actual six-point head D. Retain the literal interfaces

$$D = \{1, \ldots, 6\}, \quad \tau = \rho_3 = (6\ \beta)\delta, \quad \sigma = \rho_4, \quad H = G_{\geq 5}, \quad K = G_{>5}.$$

Here $G_{\geq 3} = \mathrm{Alt}(\Omega \setminus \{1, \ldots, 5\})$, δ is an odd involution supported outside $D \cup \{\beta\}$, and σ fixes D. The group H fixes the seven head points $D \cup \{\beta\}$. Every colour $j \geq 5$ has two edges joining distinct orbits of its full ambient omission G_j; these are called ambient eligible. The actual anchor $z_0 = \beta^\sigma$ lies outside those seven points, is fixed by K and by δ, and is not H-fixed. The degree-independent natural alternating overlap proof also gives a pure three-cycle on $\{6, \beta, z_0\}$ in $\langle \tau, \sigma \rangle$.

Choose j by the complete strong simultaneous-witness theorem. Each actual H-orbit has at most one ambient-eligible j-edge. Every witness orbit on which K is intransitive has a full local unique crossing and is fixed pointwise by δ. Suppose some witness orbit O is transitive under K. Necessarily $j \geq 6$, since an ambient 5-edge cannot have both endpoints in a transitive K-orbit. The complete whole-orbit transport supplies $p \in G_{<5}$ for which

$$T = O^p$$

is an actual K-orbit in a K-intransitive H-orbit W. The transport commutes with K, preserves the ambient witness edge, and transports the complete marked K-action. In particular O, T are disjoint and δ fixes T pointwise.

Lemma 1 (The transported action has enough points)**.** *The active K-labels on O and T form an interval $6, \ldots, x$ for some $x \geq j$. Every omission of a label in $6, \ldots, x-1$ is intransitive in this actual restricted action. Consequently*

$$|O| = |T| \geq x - 4.$$

No intersection property for the restricted action is assumed.

Proof. First apply the complete commuting-component argument at the unique ambient j-edge to the transitive K-action on O. For clarity, any component of its noncommutation diagram not containing j commutes with ρ_j, preserves O and lies in G_j. It must fix both endpoints of the unique eligible edge. A factor of a transitive commuting product which fixes a point is trivial everywhere. Thus only the component containing j can be active. The active labels form a connected interval $m, \ldots, x$, with $m \geq 6$.

If $m \geq 7$, then ρ_5 centralizes the whole K-action on O. It preserves this actual H-orbit and fixes both endpoints of its unique ambient j-edge, since $j \geq 7$. It therefore fixes all of O. The active H-interval on this witness would start at $m \geq 7$, contrary to the complete acquired first-active-label theorem. Hence $m = 6$. Notice that this argument does not assume ρ_5 fixes a general K-transitive witness.

The omission of 6 is intransitive. If $j = 6$, its two ambient witness endpoints already lie in distinct orbits of that omission. If $j \geq 7$, work on the transported orbit $T \subset W$. The generator ρ_5 fixes the transported edge endpoints, because W has its unique ambient j-edge and $\rho_5 \in G_j$ commutes with ρ_j. If $G_{\geq 7}$ were transitive on T, its centralization by ρ_5 would force ρ_5 to fix all of T. Then both K and ρ_5 would preserve T, making $W = T$, contrary to the K-intransitivity of W.

For an internal label $6 < i < x$, its omission is also intransitive. For $i = j$ this follows from ambient eligibility. Otherwise, suppose the omission were transitive. Its left and right factors commute and are both nontrivial, since all labels $6, \dots, x$ are active. The factor on the side not containing j centralizes ρ_j and fixes both endpoints of the unique ambient j-edge in O. Transitivity of the commuting product would make that factor trivial, a contradiction. This proves every asserted omission.

Put $d = x - 5$, the number of active marks. For each of the first $d - 1$ labels choose an edge joining distinct orbits of its full restricted omission. These selected edges form a forest: a cycle would join the endpoints of any one of its edges using only the other labels, contradicting eligibility. Thus the degree is at least d. If it were exactly d, the forest would be a spanning tree. Deleting one selected edge divides the tree into two connected sets. Its intransitive full omission contains every other selected edge, so its orbits are exactly those two sets. The last marked generator belongs to every one of these omissions and therefore preserves both sides of every tree cut. Tree cuts distinguish every pair of vertices, so it fixes every vertex. This contradicts its nonidentity. When $d = 1$, degree one likewise makes the sole mark trivial. Therefore the degree is at least $d + 1 = x - 4$, as claimed. □

Theorem 2. *With the actual transitive witness and transport above, the last active label x satisfies*

$$x < r - 1, \qquad |\operatorname{Fix}(G_{>x})| \geq 2x + 1.$$

Proof. The eight points $D \cup \{\beta, z_0\}$ and the two nontrivial K-orbits O, T are disjoint. They are all fixed by every mark after x. The preceding lemma gives at least

$$8 + 2(x - 4) = 2x \tag{1}$$

fixed points.

Suppose first that $x < r - 1$ but the asserted stronger bound fails. Then the fixed set F consists exactly of these eight points and the two K-orbits, with $|O| = |T| = x - 4$. The involution δ centralizes K and $G_{>x}$, and fixes the eight points and T pointwise. It preserves F, so it must preserve O as well. Its restriction to O is the restriction of the actual $\tau \in G_j$. It commutes with ρ_j and preserves the unique ambient j-edge there. Since its endpoints belong to different G_j-orbits, each endpoint is fixed. Centralization of the transitive K-action now makes δ fix all of O.

Thus δ fixes F pointwise. The set F is K-invariant by its displayed decomposition. It is preserved by ρ_5, which commutes with $G_{>x}$ because $x \geq 6$, and by σ, which also centralizes that suffix. The first three marks preserve D, and $\tau = (6\ \beta)\delta$ preserves F. Hence every generator of G preserves F. Transitivity would force $F = \Omega$, but then the odd involution δ would be identity. This contradiction proves the stronger bound whenever x is nonterminal.

It remains to exclude $x = r - 1$. Write $N = |O| = |T|$. The head and these two orbits occupy at least $8 + 2(r - 5) = 2r - 2$ points, leaving at most three. If δ preserves O, the same unique-edge argument fixes it pointwise. The odd involution δ would then be supported on at most three points, hence be a literal transposition. This is excluded by the complete acquired exterior-pair obstruction.

Otherwise δ sends O to a distinct orbit V of H, because it centralizes H. It is also an actual K-orbit, equivariantly isomorphic to O. It is disjoint from T, which δ fixes pointwise, and from the eight fixed points. Consequently $N \leq 3$. Since $N \geq r - 5$ and $x = r - 1 \geq 6$, only the following cases can occur. For $r = 7$, necessarily $N = 2$, since three copies of degree three would not fit. The head and the three copies occupy fourteen of the fifteen points. The involution δ exchanges O, V by two transpositions, fixes T and the eight head points, and fixes the one remaining point. It is even, a contradiction.

For $r = 8$, necessarily $N = 3$, and the eight head points and the three copies exhaust all seventeen points. On each copy the two active involutions ρ_6, ρ_7 are distinct transpositions generating natural S_3. Their equivariant identifications make K diagonal on three natural rows. The centralizer of natural S_3 on one row is trivial, and there is a unique equivariant bijection between

any two rows. Thus every permutation centralizing K acts on the row union solely by permuting the three rows.

The row action of δ exchanges O, V and fixes T. The row action of σ must exchange T with one of O, V. Otherwise $O \cup V$ would be invariant under every generator: H preserves its two actual orbits, τ exchanges them through δ, σ preserves their union, and the first three marks fix it. This contradicts transitivity. Thus the two row involutions are distinct transpositions. Their product is a three-cycle on the rows.

On the fixed eight-point set, τ and σ are respectively $(6\ \beta)$ and $(\beta\ z_0)$, with the other five points fixed. Their product is likewise a three-cycle. Hence the dihedral group $\langle \tau, \sigma \rangle$ has its nonidentity rotations nontrivial both on the three-point core and on the row union. Its reflections have nontrivial row transpositions. It cannot contain a pure three-cycle on $\{6, \beta, z_0\}$, contrary to the acquired actual overlap theorem. This excludes the final terminal case and proves the theorem. □

Attribution and boundary. The strong witness colour, whole-orbit transport, actual head anchor and pure core cycle are acquired. The commuting-factor proof, omission forest and complete fixed-set argument supply the new separator. The terminal three-row argument reuses the acquired odd-degree row mechanism with its actual hypotheses verified here; no earlier row-assignment theorem is invoked with changed assumptions. The second witness need not be locally unique in its restricted K-action, and ρ_5 is not assumed to fix it. Full global source routing and attachments remain. No whole-original, question, numerical or formal credit is claimed.